Tracking the Evolution of E-Commerce Websites in Portugal

Exploração de Dados

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Context



Porpuse

- Study how e-commerce websites in Portugal have changed in terms of offerings, product availability and price.



Data

- Gather product information of popular e-commerce sites in Portugal over time from *Arquivo.pt*.



Objectives

- Understand the evolution of e-commerce websites in Portugal
- Analyze changes in product availability, pricing, and diversity over time.

Data Acquisition



Source

- The data was gathered using the *Arquivo.pt* **API** to fetch archived versions of the e-commerce websites.



Methodology

- Defined search terms for categories.
- Selected few majors websites.
- Implemented a script to iteratively request data from the Arquivo.pt API while handling pagination.

What is new since the last presentation?

Web Scraping

Objective

- Extract *title*, *price* and *category* from the product in the page retrieved.

Implementation

- Read data from a *CSV* file previously stored from the request to the *Arquivo API*.
- Extract *title*, *price* and *category* using BeautifulSoup.
- For each site there is a list of specific classes to be able to extract data from different sites.
- Data Validation
- Store results in a new CSV file.

Data Treatment

Price Format Correction

- Identification of poorly formatted prices (e.g.: 1,129,99) and converting to the correct format.

{category}

Removing Invalid Values

- Exclusion of prices in invalid data is present and elimination of non-numeric characters and unnecessary spaces.

Validação de Dados

- Removal of rows with empty prices or NaN.

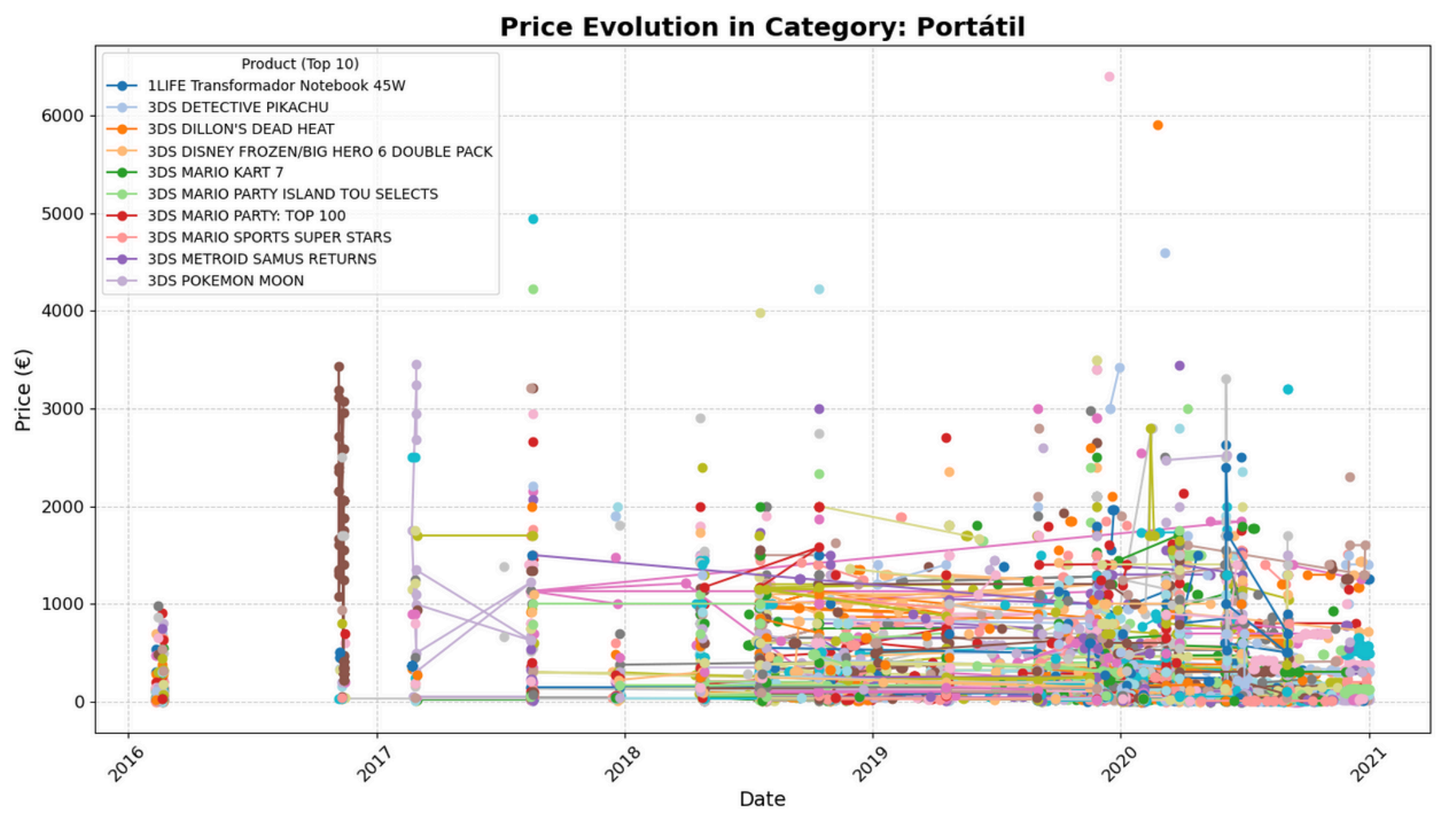
Data Treatment



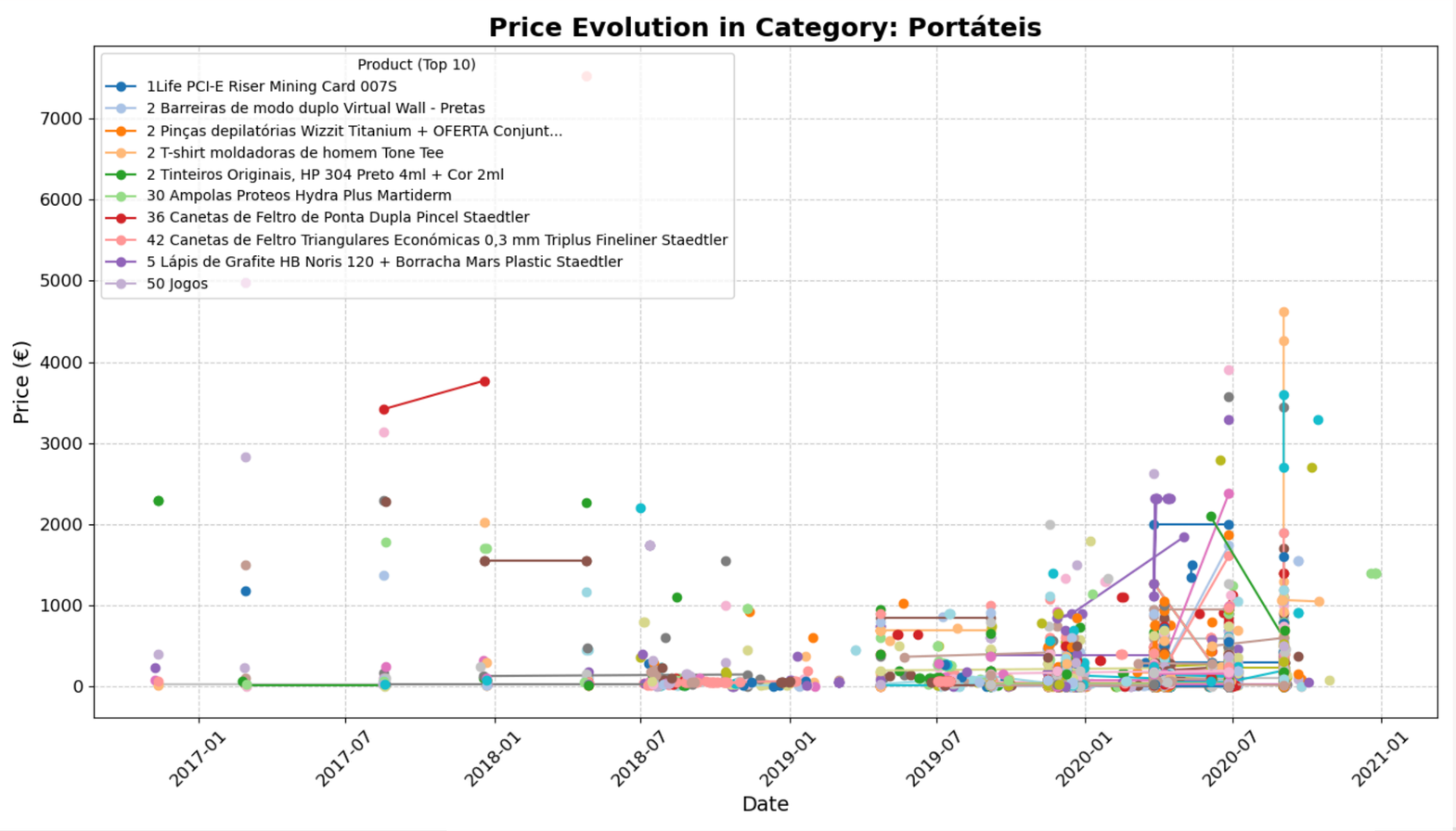
Implementation

- Reads the CSV files with the results from the Web Scraping
- Application of Cleaning Rules
- Saves the Corrected Data in CSV files.

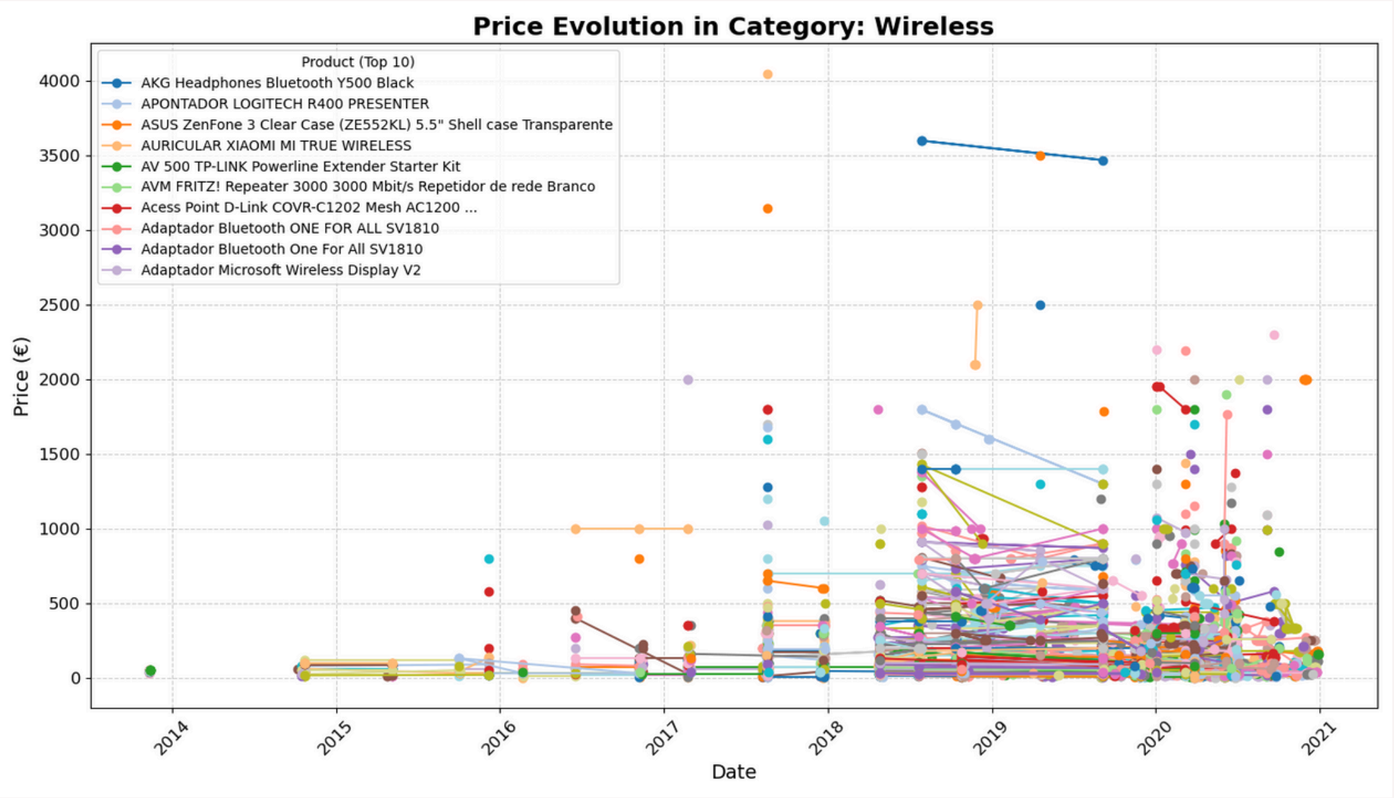
Visualizations



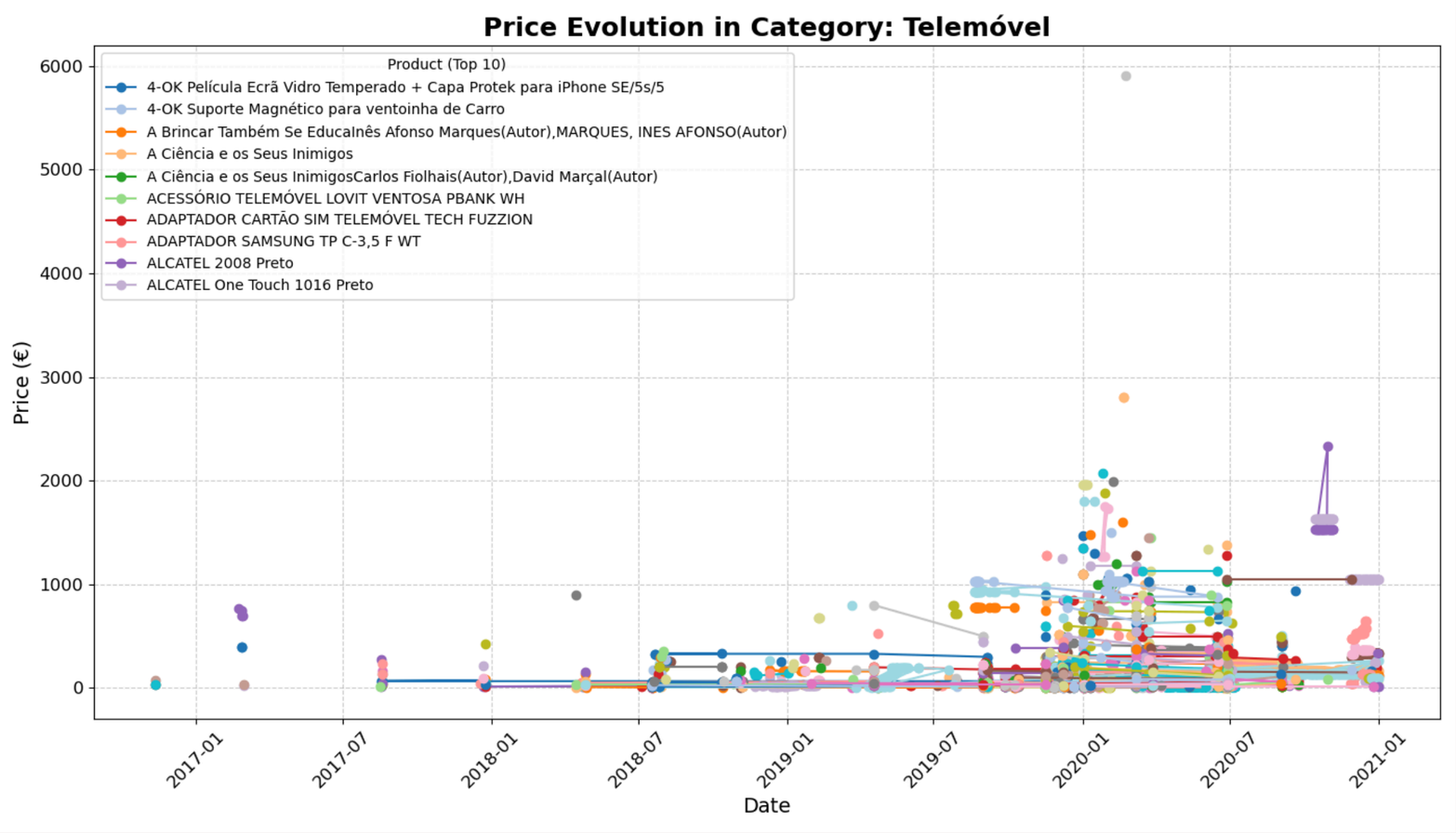
Visualizations



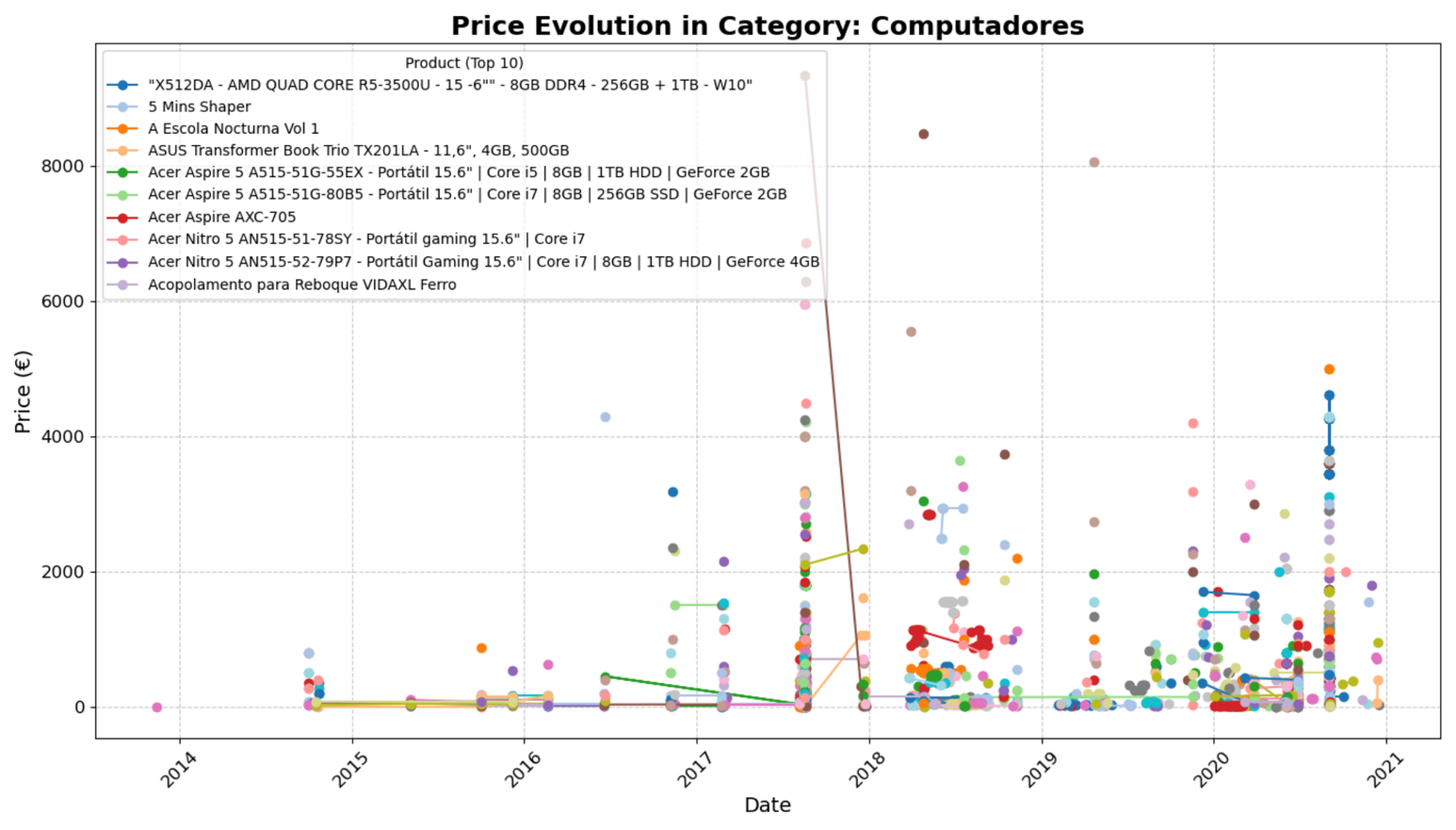
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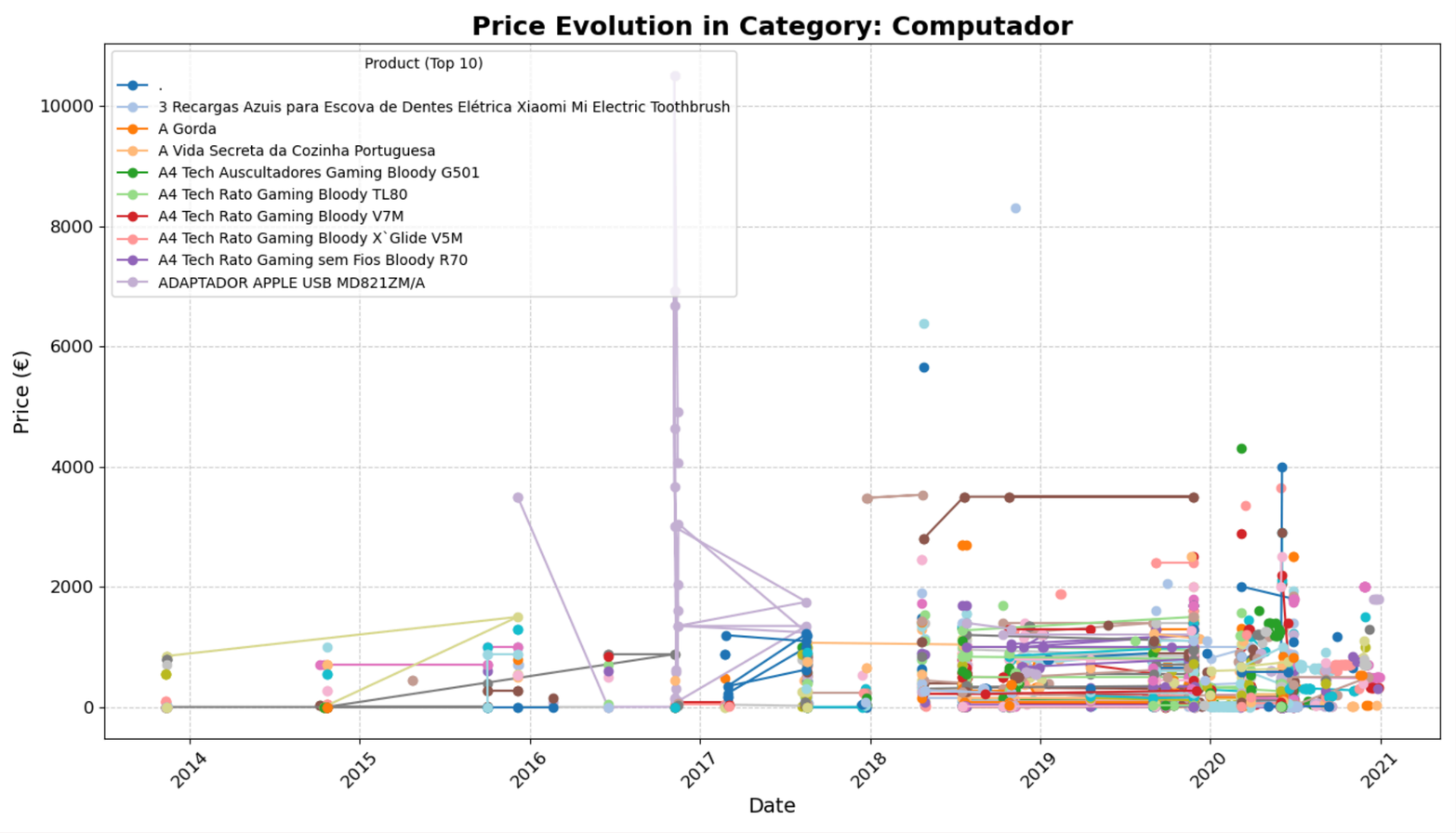
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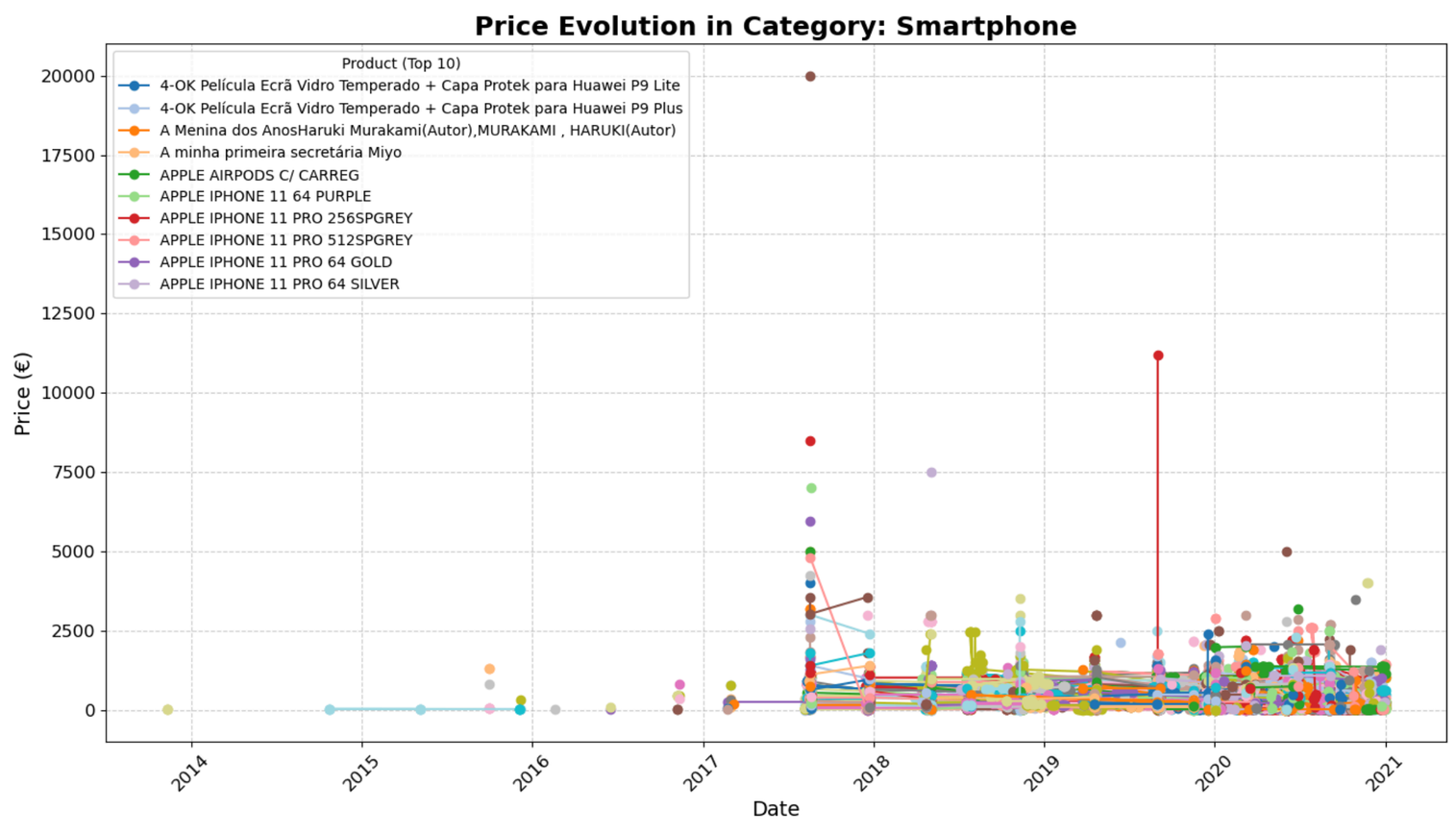
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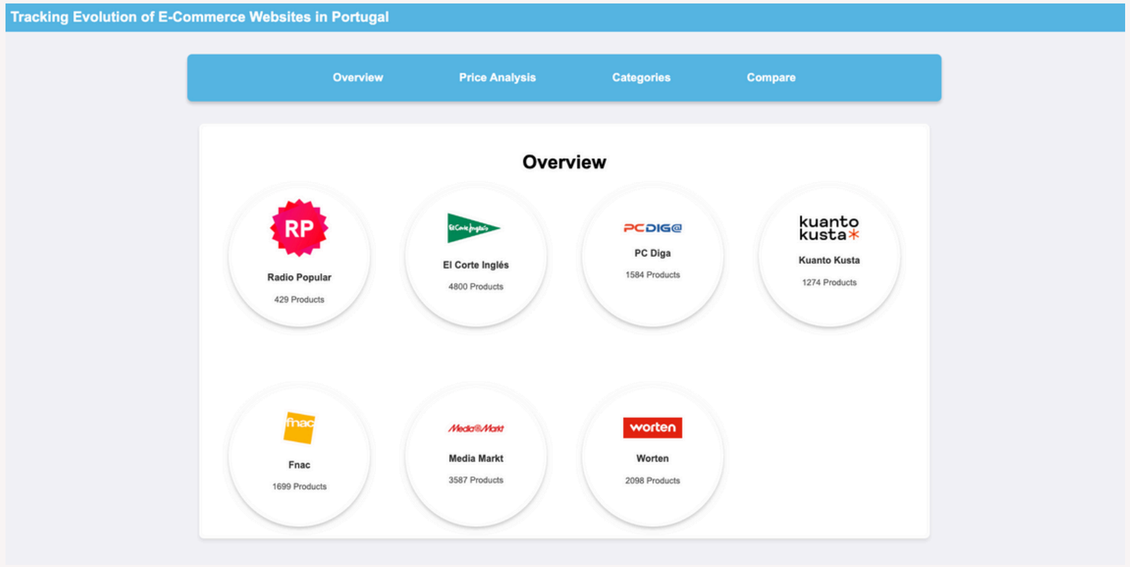
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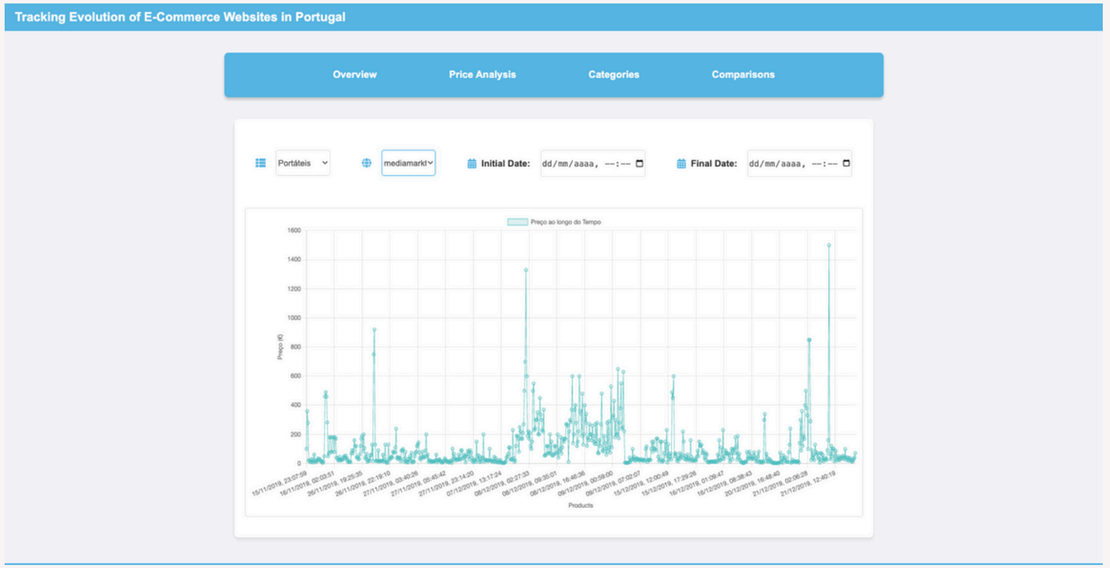


Visualizations



User Interface





THANKS!

Do you have any questions?

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